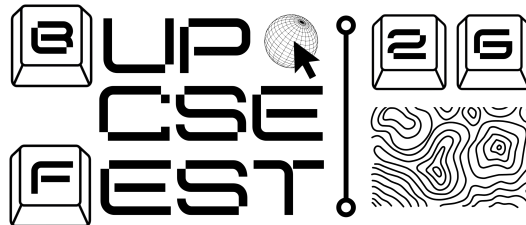


BANGLADESH UNIVERSITY OF PROFESSIONALS

Mirpur Cantonment, Dhaka - 1216



BUP CSE FEST 2026

HACKATHON

*In association with **Poridhi***



Official Rulebook

All participating teams of the **Hackathon** will receive **Poridhi's Puku Editor, CLI and Starter Lab Access** for unlimited AI coding assistant & support during the hackathon.

Winner Team will additionally receive exclusive **Awards from Poridhi** as part of their prize package.

fest.bupcopc.tech

Contents

1	The Event	3
2	Competition Structure	3
2.1	Phase 1 — Online Preliminary Round	3
2.2	Phase 2 — Final 8-Hour On-Site Hackathon	3
3	Judging Criteria	3
4	Rules & Guidelines	4
4.1	Team Composition	4
4.2	Institution Requirements	4
4.3	Project Rules	4
5	Facilities	5
6	Segment Description	5
6.1	What is Allowed	5
6.2	What is NOT Allowed	5
6.3	Sample Problem Scenario	6
7	Submission Requirements	7
7.1	Presentation Format (On-site Finals)	8
8	Code of Conduct	8
8.1	Expected Behaviour	8
8.2	Prohibited Conduct	8
8.3	Procedure & Appeals	8

1. The Event

The AI Hackathon is the flagship challenge of **BUP CSE Fest 2026**, hosted by the BUP Computer Programming Club under the Department of Computer Science and Engineering at Bangladesh University of Professionals (BUP), Dhaka. The event gathers builders and creative problem-solvers from across the country to develop practical, intelligent tools that bridge artificial intelligence with real-world APIs.

This year's event is supported by **Poridhi** as the official hackathon partner, providing participants with access to the **Puku Editor** and **CLI** for unlimited AI-assisted coding.

Teams have **8 continuous hours** to take an idea from scratch to a working full-stack build—whether targeted at **Web, Android, or both**. The core problem statement will be revealed live at the Opening Ceremony. Solutions do not need to be production-ready, but they must be functional, demonstrable, and directly aligned with the challenge prompt.

Mentors and challenge designers from both industry and academia will be on-site around the clock to offer guidance. Ultimately, the judges are looking for working prototypes that showcase solid technical intuition, efficiency, scalable thinking, and a genuinely creative approach to solving problems with AI.

2. Competition Structure

2.1 Phase 1 — Online Preliminary Round

A 3-4 hour online qualification round where teams solve screening challenges to advance to the main event.

- Teams are evaluated on problem-solving ability, technical reasoning, and solution quality.
- The **top 50 teams** will be selected to participate in the main on-site hackathon.

2.2 Phase 2 — Final 8-Hour On-Site Hackathon

An intensive 8-hour on-site hackathon for all qualifying teams, held at the BUP campus.

- The problem statement/scenario is announced at the Opening Ceremony.
- Teams must develop their entire solution from scratch within the 8-hour window.
- Mentors from industry and academia are available on-site throughout the event.
- A minimum of 10 teams will be selected for the final presentation round.

3. Judging Criteria

Projects will be evaluated based on the following criteria:

- Problem Understanding & Impact
- Functionality & Completeness
- AI/API Integration
- Technical Design & Innovation
- Presentation & Live Demo

4. Rules & Guidelines

4.1 Team Composition

- Each team must consist of **2–4 members**.
- All participants must be currently pursuing **undergraduate** degrees.
- **Cross-university teams are allowed** — members from different universities may form a team.
- No participant may be a member of more than one team.
- All participants must be physically present at the venue throughout the main hackathon. Remote participation will not be allowed.

4.2 Institution Requirements

- Participating institutions must be registered under the University Grants Commission (UGC) of Bangladesh.
- Participation is limited to students of Bangladeshi institutions only.

4.3 Project Rules

- Participants must create a new GitHub repository immediately after the Opening Ceremony/Question reveal and submit it at the end of the competition (both preliminary and finals).
- The GitHub repository must remain private for the duration of the event and must be made public after the submission deadline for evaluation.
- The project must be developed entirely on the day(s) of the hackathon. Any project developed fully or partially before the event will not be accepted.
- Teams may use publicly available open-source libraries, frameworks, APIs, and SDKs. All must be credited in the project README.
- Use of AI-assisted tools (e.g., GitHub Copilot, ChatGPT) is permitted. Core architecture and logic should be the team's own work.
- Teams may bring pre-designed wireframes or planning documents. No pre-written source code may be brought into the venue.
- A scenario and set of tasks will be presented at the Opening Ceremony. Teams are free to use any language, tool, or framework as long as they fulfil the task requirements.

IMPORTANT

Any unfair attempts, misbehaviour, or breach of rules will result in immediate disqualification. The organizers reserve the right to make final decisions on all matters.

5. Facilities

What We Provide

- **Poridhi Puku Editor and CLI — Unlimited AI Coding Assistant:** Poridhi will provide participants with access to the Puku Editor and CLI for unlimited AI coding assistant support during the hackathon.
- **Internet:** Wi-Fi connectivity will be provided throughout the venue for the duration of the hackathon.
- **Food:** Meals and refreshments will be provided for all participants for the duration of the event.
- **Security:** The venue will be secured throughout the event by the organizing team and campus security.

What Participants Should Bring

- **Laptop(s)** — each team member should bring a personal laptop.
- **Chargers** — laptop chargers and any device-specific chargers.
- **Multiplugs / Power Strips** — to manage multiple devices at a single workstation.
- **Mobile Data (Backup)** — although Wi-Fi will be provided, participants are strongly recommended to carry mobile data as a backup due to potential infrastructural constraints.
- Any other necessary personal or development devices (external keyboards, mice, etc.).

NOTE

The organizing committee will do its best to ensure stable internet connectivity. However, due to the high number of concurrent users and possible infrastructure limitations, mobile data as a backup is strongly advised to avoid disruptions during critical development periods.

6. Segment Description

Participants will build full-stack functional solutions — **Web or Android or Both** — that combine Artificial Intelligence with API-based integrations to address a real-world problem scenario announced on the day. Teams should aim to demonstrate process understanding, efficiency, scalability, and creative AI integration.

Creative and effective use of AI services or custom models that align well with the problem requirements will be highly rewarded by the judging panel.

6.1 What is Allowed

- Any publicly available APIs, SDKs, or your own trained AI/ML models.
- Pre-trained models accessed via API (e.g., OpenAI, Hugging Face, Azure Cognitive Services, Google Cloud AI, Anthropic Claude API).
- Any programming language, web framework, or mobile development toolkit.
- Cloud services (AWS, Google Cloud, Azure, etc.).

6.2 What is NOT Allowed

- Pre-written source code of any kind brought into the venue.
- Projects developed, even partially, before the hackathon begins.

6.3 Sample Problem Scenario

SAMPLE PROBLEM SCENARIO

Scenario: AI-Driven Climate-Resilient Agricultural Advisory & Crop Health Platform

Background

Agriculture sustains over 40% of Bangladesh's national workforce, yet smallholder farmers routinely lose 20–30% of seasonal yields to preventable pest infestations, leaf blights, and sudden climatic shifts. Sub-Assistant Agriculture Officers (SAAOs) are tasked with overseeing thousands of farming families across entire unions, making timely on-field inspection and specialized diagnostics practically impossible. Furthermore, historical farm logbooks, soil test reports, fertilizer application charts, and localized market price receipts remain unorganized on paper, cutting farmers off from data-driven decisions.

The Challenge

You are tasked with engineering a full-stack **AI-Powered Agro-Advisory & Field Intelligence System** that enables grassroots field extension workers and farmers to diagnose crop diseases on-site, analyze local micro-climate vulnerabilities, predict fair market selling prices, and receive immediate actionable interventions in both Bengali and English.

Task 1 — Voice-Guided Farmer Query Intake & Intent Extraction

A field agent or farmer should be able to dictate current crop symptoms, cultivation stage, and observed anomalies in natural spoken Bengali or English. Use a Speech-to-Text API (e.g., Whisper API, Google Cloud Speech-to-Text, AssemblyAI) to transcribe the audio stream. Parse and normalize the raw transcript using an NLP/Translation pipeline into a structured JSON schema containing crop type, estimated planting date, observed damage description, and geographic union.

Task 2 — Visual Crop Disease Detection & Damage Severity Estimation

Users capture or upload field photographs of infected leaves, stems, or fruits. Implement a Computer Vision pipeline (e.g., custom fine-tuned YOLO/PyTorch model, Google Cloud Vision, Roboflow, or multimodal vision LLMs) to identify the specific pathogen (e.g., Rice Blast, Potato Late Blight, Tomato Leaf Curl Virus). The model must pinpoint infected regions, compute a visible surface damage percentage, and output an infection severity classification (**Mild / Moderate / Severe / Critical**).

Task 3 — Multimodal Agronomic Reasoning & Treatment Engine

Pass the structured symptom profile, computer vision outputs, and real-time hyperlocal weather data (fetched via an external weather API) into an LLM-based agronomic reasoning module (e.g., Claude, OpenAI, Gemini) with contextual system prompts. The engine must:

- Explain the probable root cause, accounting for current environmental factors (e.g., high humidity triggering fungal spore multiplication).
- Generate a tiered intervention plan detailing both non-chemical organic controls and exact chemical pesticide/fungicide dosages.
- Issue safety precautions, mandatory pre-harvest intervals (PHI), and a localized weather-adjusted spraying schedule (e.g., abort spraying if rain is forecast within 6 hours).

Task 4 — Yield Risk & Market Price Anomaly Detection

The user logs farm inputs, harvested batch volumes, and offered prices from local middlemen (*farias/arotdars*). Build an analytical/ML anomaly detection module (e.g., Isolation Forest, Z-score modeling, or time-series regression) to cross-reference entries against historical wholesale commodity rates. The model should flag predatory pricing undercuts, calculate current price volatility, and estimate an optimal 7-day selling window.

Task 5 — Bengali Audio Advisory & Digital Crop Passport Generation

Upon completing diagnosis and market evaluation, the system should:

- Synthesize a clear, spoken voice briefing of critical treatment steps and weather cautions using a Text-to-Speech API (e.g., ElevenLabs, Azure Neural TTS, Google Cloud TTS) in conversational Bengali.
- Generate a printable/downloadable digital summary report (“Field Health Card”) containing the captured image with bounding overlays, detected pathology, dosage directions, risk score, and market price benchmarks suitable for sharing via WhatsApp or SMS.

NOTE

This is a sample scenario only. It illustrates the format, complexity, and depth of AI integration expected. The actual problem statement will be different and will be announced at the Opening Ceremony. Teams should use this sample to calibrate their preparation and API setup — not to pre-build a solution.

7. Submission Requirements

#	Deliverable	Details
1	GitHub Repository	Public repo with all code written during the hackathon. Include a <code>README.md</code> with setup, run instructions, and tech stack. Final evaluation uses the last commit before the deadline.
2	Presentation Slides (Finals)	PDF or PPTX, maximum 15 slides. Cover: problem, solution, architecture, tech used, challenges, and future roadmap.
3	Live Demo	Mandatory working prototype demonstration before the judging panel. The presentation must primarily focus on the live demo.
4	Project Documentation	Optional but recommended: testing notes, API integration guide, deployment steps.

Every team must submit all required deliverables through the official submission portal. Submissions after the deadline will not be considered under any circumstances.

IMPORTANT

HARD DEADLINE: Teams missing the submission deadline forfeit their right to present and will be excluded from judging.

7.1 Presentation Format (On-site Finals)

- Each team is given **10 minutes total**: 7 minutes for the pitch and live demo + 3 minutes for judge Q&A.
- A minimum of 10 teams will be selected for the final presentation round.
- Presentation order is determined by a random draw after the submission deadline.
- Teams must use the provided projector/screen. Personal external displays are not permitted.
- The presentation should primarily focus on the live demonstration of the working prototype.

8. Code of Conduct

BUP CSE Fest 2026 is committed to providing a safe, inclusive, and respectful environment for all participants, judges, mentors, volunteers, and guests. Harassment, discrimination, and disrespectful behaviour of any kind will not be tolerated.

8.1 Expected Behaviour

- Treat all individuals with respect and professionalism at all times.
- Foster a collaborative spirit — share knowledge, support fellow participants, and maintain sportsmanship throughout.
- Keep workstation areas tidy and respect all shared spaces.
- Comply with all instructions from the organizing committee and university authorities.

8.2 Prohibited Conduct

- Harassment, intimidation, bullying, or discrimination of any kind against any person.
- Use of offensive, derogatory, or discriminatory language in person or within project submissions.
- Consumption of alcohol, tobacco, or illegal substances on BUP campus premises.
- Any form of plagiarism, academic dishonesty, or misrepresentation of work.

8.3 Procedure & Appeals

If a violation is suspected, the concerned team will be formally notified and given a reasonable opportunity to respond. Disqualified teams forfeit all prizes and recognition.

Teams may submit a written appeal within 1 hour of the disqualification notice. The Organizing Committee's decision is final and binding.

BUP CoPC, Department of Computer Science & Engineering
Bangladesh University of Professionals, Dhaka, Bangladesh

fest.bupcopc.tech